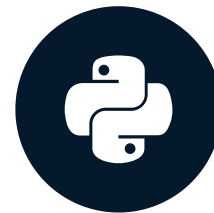


Introduction

PYTHON FOR R USERS



Daniel Chen
Instructor

The basics

- Variables
- Assignment
- Strings
- Lists
- Dictionaries

Programming fundamentals

- Control flow (if-statements)
- loops
- Functions

Pandas

- Selecting and subsetting data
- Data types
- Data manipulation and processing techniques

Plotting

- pandas
- matplotlib
- seaborn

Putting it all together

- NYC Flights 2013 Dataset
- Planes
- Flight Delays

Python data types

R

- `numeric` : integers, floating point numbers
- `logical` : `TRUE` or `FALSE` values
- `character` : strings

Python

- `int` : integers
- `float` : floating point numbers
- `bool` : `True` or `False` values
- `str` : strings

Assignment

R

```
x <- 3  
  
y = 0.5  
  
z <- TRUE
```

Python

```
x = 3  
  
y = 0.5  
  
z = True
```

Print statements

R

```
print('R is awesome!')
```

```
"R is awesome!"
```

Python

```
print('Python too!')
```

```
Python too!
```

What's your data type?

R

```
x <- 3  
  
class(x)
```

```
"numeric"
```

Python

```
x = 3  
  
type(x)
```

```
int
```

```
y = 0.5  
  
type(y)
```

```
float
```

R

```
1 + 1
```

```
2
```

```
'1' + '1'
```

```
Error in "1" + "1" : non-numeric argument to binary operator
```

```
'1' * 5
```

```
Error in "1" * 5 : non-numeric argument to binary operator
```

```
'1' '1'
```

```
Error: unexpected string constant in "'1' '1'"
```

Python

```
1 + 1
```

```
2
```

```
'1' + '1'
```

```
'11'
```

```
'1' * 5
```

```
'11111'
```

```
'1' '1'
```

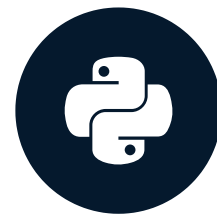
```
'11'
```

Let's practice!

PYTHON FOR R USERS

Containers - lists and dictionaries

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Lists

- Similar to the R vector and list
- Holds heterogeneous information (e.g.,
`[3, 2.718, True, 'hello']`)

Creating lists

R

```
l <- list(1, "2", TRUE)
l
```

```
[[1]]
[1] 1

[[2]]
[1] "2"

[[3]]
[1] TRUE
```

Python

```
l = [1, '2', True]
l
```

```
[1, '2', True]
```

Python is 0-indexed

Python is a 0-indexed language

Subsetting lists

R

Get the 1st element from R list

```
l[[1]] # use 1
```

```
1
```

Python

Get the 1st element from
Python list

```
l[0] # use 0
```

```
1
```

Negative indices

Negative indices start counting from the end

Negative indices

R

```
l <- list(1, "2", TRUE)  
l[-1]
```

```
[[1]]  
[1] "2"  
  
[[2]]  
[1] TRUE
```

Python

```
l = [1, "2", True]  
l[-1]
```

```
True
```

Left inclusive right exclusive

Left inclusive, right exclusive

Think fence posts

index	0	1	2	3	4	5
	element 0		element 1		element 2	
	element 3		element 4			

Slicing part 1

Python

```
l = [0, 1, 2, 3, 4]
```

```
l[0:3]
```

```
[0, 1, 2]
```

```
l[0:5:2]
```

```
[0, 2, 4]
```

Slicing part 2

Implicit

```
l[:3]
```

```
[0, 1, 2]
```

```
l[1:]
```

```
[1, 2, 3, 4]
```

```
l[::2]
```

```
[0, 2, 4]
```

Explicit

```
l[0:3]
```

```
[0, 1, 2]
```

```
l[1:5]
```

```
[1, 2, 3, 4]
```

```
l[0:5:2]
```

```
[0, 2, 4]
```

Dictionaries

- Lists are unlabeled
- Dictionaries provide key:value pairs

Python: Creating and working with dictionaries

```
d = {'int_value':3, 'bool_value':False, 'str_value':'hello'}
```

```
d
```

```
{'bool_value': False, 'int_value': 3, 'str_value': 'hello'}
```

```
d['str_value']
```

```
'hello'
```

Finding the length

```
l = [0, 1, 2, 3, 4]
d = {'int_value':3, 'bool_value':False, 'str_value':'hello'}
```

```
len(l)
```

```
5
```

```
len(d)
```

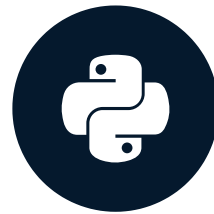
```
3
```

Let's practice!

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Functions, methods, and libraries

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Calling functions

- R and Python functions work the same
- Python: Functions and Methods

Methods vs functions

- Python is object oriented
 - Attributes
 - Methods
- Methods are functions that an **object can call** on itself
- Functions are **called on an object**

```
l = [0, 1, 2, 3, 4]
len(l)    # call a function
l.len()   # not a method
```

Periods have a special meaning in Python

Periods have very specific meanings and uses

Append to list

```
l = [1, "2", True]
```

```
l.append('appended value')
```

```
l
```

```
[1, '2', True, 'appended value']
```

Update a dictionary

```
d = {'int_value':3, 'bool_value':False, 'str_value':'hello'}
```

```
d.update({'str_value':'new_value', 'new_key':'new_value'})
```

```
d
```

```
{'bool_value': False,  
 'int_value': 3,  
 'new_key': 'new_value',  
 'str_value': 'new_value'}
```

Methods are specific to the object

```
d.append('hello')
```

```
-----  
AttributeError                                Traceback (most recent call last)  
<ipython-input-4-42922a99ec56> in <module>()  
<hr />-> 1 d.append('hello')  
  
AttributeError: 'dict' object has no attribute 'append'
```

Libraries

- Libraries provide more functionality
- Arrays and dataframes are not built into Python
- Arrays and matrices come from `numpy`
- Dataframes come from `pandas`

Numpy ndarray

R

```
library(readr)
dat <- read_csv('my_file.csv')
```

Python

```
import numpy
arr = numpy.loadtxt('my_file.csv', delimiter=',')

import numpy as np
arr = np.loadtxt('my_file.csv', delimiter=',')
```

Pandas DataFrame

```
import pandas as pd
```

```
df = pd.read_csv('my_file.csv')  
df.head()
```

Let's practice!

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